

Music Player

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Description:

A music player that reads audio files from a MicroSD card and plays them through a speaker. It features an OLED display that shows the current track name alongside a real-time audio visualization. Additionally, it includes an infrared remote control, allowing the user to play, pause, and skip songs from a distance without using the physical buttons on the device.

List of hardware components:

- MicroSD card reader module (with SPI interface) (6-pin)
 - MAX98357A I2S Audio Amplifier Module
 - Miniature speaker 3W (4 Ohm or 8 Ohm)
 - OLED display 0.96 inch SSD1306 (I2C) (4 pins)
 - Rotary Encoder Module (ex: KY-040)
 - 2x Touch buttons (Push buttons - 6x6mm)
 - Breadboard (high 830 points)
 - A Father-Father set
 - A Mom-Dad set
 - IR remote control kit with receiver (38kHz)
- + CR2025 battery
- + MicroSD card

List of software libraries:

- embassy-stm32
- embassy-executor
- embassy-time
- embassy-sync
- defmt
- defmt-rtt
- panic-probe
- embedded-sdmmc
- microfft
- ssd1306
- embedded-graphics
- infrared
- rotary-encoder-hal